Ankit Gautam

Mechanical Engineer | Fluid Dynamics | R&D

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Professional Summary

Mechanical Engineer and PhD researcher with 4+ years of industrial mechanical engineering experience and advanced research experience in fluid dynamics, experimental mechanics, computational modeling, and mechanical design.

Core Engineering Skills

- **Fluid Dynamics & Modeling:** Fluid Mechanics, Computational Fluid Dynamics (CFD), Flow Modeling, Numerical Analysis, Physics-Based Modeling, Turbulence, Flow Measurements, Model Validation
- **Scientific Computing & Programming:** Python, MATLAB, C, LabVIEW, Scientific Computing, Engineering Tool Development, Data Processing, Automation, Numerical Methods, Signal Processing
- **Simulation & Analysis:** ANSYS, OpenFOAM, Finite Element Analysis (FEA), Computational Modeling, Design Analysis, Quantitative Analysis, Simulation Post-Processing, Data Visualization
- **Experimental & Validation:** Experimental Design, PIV, Instrumentation, Data Acquisition, Laboratory Testing, Performance Characterization, Measurement Analysis, Experimental Validation
- **Mechanical Engineering:** Mechanical System Design, Component Design, Structural Analysis, Prototype Development, Design Validation, Engineering Drawings
- **Engineering Tools:** SolidWorks, CATIA, AutoCAD, LaVision DaVis, Linux/Windows Engineering Workflows, Technical Documentation

Experience

Doctoral Researcher
Experimental Fluid Dynamics Labrotory at Utah State University — *Logan, UT* — *2022 – Present*

- Built a custom wind-tunnel facility with active grid and multi-panel synthetic-jet array for controlled generation of turbulence. Led PIV measurements and multi-point instrumentation.
- Performed advanced analysis including CFD simulations and Proper Orthogonal Decomposition (POD) analysis to quantify the impact of coherent structures on turbulence. Published peer-reviewed results.

Mechanical Design Engineer
Units Engineering Consultancy (Visit) — *Kathmandu, Nepal* — *2018 – 2022*

- Applied CFD (ANSYS Fluent) and FEA to validate flow performance and structural integrity of mechanical components for multiple commercial hydropower projects under IS, IEC, and ISO standards.
- Generated detailed engineering drawings and supported pre-commissioning inspections for systems that successfully entered operation or construction.

Research Assistant
Center for Energy Studies, Tribhuvan University (Visit) — *Lalitpur, Nepal* — *2017 – 2018*

- Engineered mechanical components for a novel 1 kW gravitational water vortex hydropower prototype using analytical and computational fluid models.
- Supervised prototype fabrication, assembly, structural testing, and operational performance benchmarking.

Education

Ph.D., Mechanical Engineering – Utah State University, Logan, UT — Expected May 2027
M.S., Mechanical Engineering – Tribhuvan University, Nepal — 2021
B.S., Mechanical Engineering – Tribhuvan University, Nepal — 2016

Selected Projects

Turbulence Generation and Characterization (Publication-1, Publication-2)
Designed and controlled a custom experimental facility to study coherent structures and turbulence. Combined PIV experiments with CFD and POD analysis; results published in reputed journals.
Tools: SolidWorks • PIV (LaVision DaVis) • LabVIEW • Python • MATLAB • OpenFOAM • ANSYS

Gravitational Water Vortex Power Plant Prototype (Publication-3, Publication-4, Publication-5)
Engineered mechanical components and runner designs for a novel low-head hydropower prototype. Supervised fabrication, assembly, structural testing, and performance benchmarking using combined analytical and computational methods.
Tools: CATIA • ANSYS Fluent • FEA • CFD • Prototype Development

Flow-Induced Vibration Energy Harvester (Master's Thesis - Publication-6)
Numerically investigated two-degree-of-freedom galloping-based piezoelectric energy harvesting using fluid-structure interaction (FSI) modeling in ANSYS.
Tools: ANSYS Fluent/Mechanical • FEA • CFD • FSI

Commercial Hydropower Mechanical Systems
Designed mechanical components and systems for multiple commercial hydropower projects using CAD, FEA, and CFD. Delivered engineering drawings and supported successful commissioning under international design codes. (Visit)
Tools: SolidWorks • ANSYS Fluent & Mechanical • Engineering Drawings • FEA • CFD

ABU Robocon – Mechanical Design Lead (International Competition) (Visit)
Designed and built mechanical subsystems for robotics platforms competing in ABU Asia-Pacific Robot Contest (2014–2016). Represented Nepal and earned multiple awards, including Best Engineering and Panasonic Awards.(Visit)
Tools: SolidWorks • Mechanical Design • FEA • Rapid Prototyping

Selected Publications

- A. Gautam and T. Berk (2026). "Quantifying the impact of coherent structures on the turbulent kinetic energy decay rate: A Proper Orthogonal Decomposition approach ." *Physical Review Fluids, 11(4), 044906..*
- A. Gautam and T. Berk (2026). "Turbulence generation in a two-panel synthetic-jet facility with multi-variable randomisation and rotation ." *Flow, 6, p. E31.*
- T. R. Bajracharya, et al. including A. Gautam (2020). "Effects of geometrical parameters in gravitational water vortex turbines with conical basin ." *Journal of Renewable Energy, Article 5373784..*
- A. Gautam, A. Sapkota, et al. (2016). "Study on Effect of Adding Booster Runner in Conical Basin: Gravitational Water Vortex Power Plant: A Numerical and Experimental Approach ." *Proceedings of IOE Graduate Conference 4 (ISSN: 2350-8914), 107–113.*
- A. Gautam and T. Berk (2026). "Influence of Coherent Structures on TKE Decay ." *14th International Symposium on Turbulence and Shear Flow Phenomena, Heidelberg, Germany..*
- A. Sapkota, A. Gautam, et al. (2021). "Development of a 1 kW Gravitational Water Vortex Hydropower Plant Prototype ." *IJREWHS 2021.*
- A. Gautam, M.C. Luintel (2021). "Study of a flow-induced-vibration energy harvester: A case of two-degree-of-freedom galloping-based scheme ." *10th IOE Graduate Conference 10 (2350-8914), 175-182.*

Conference Presentation

- A. Gautam and T. Berk (2026, July). "Influence of Coherent Structures on TKE Decay." 14th International Symposium on Turbulence and Shear Flow Phenomena, Heidelberg, Germany.
- A. Gautam and T. Berk (2024, November). "Comparison of the homogeneous turbulence created by a turbulence box to that of an active grid." APS Division of Fluid Dynamics Meeting Abstracts, J37.001.
- A. Gautam and M. C. Luintel (2021). "Study of a flow-induced-vibration energy harvester: A case of two-degree-of-freedom galloping-based scheme." 10th IOE Graduate Conference.
- A. Gautam, A. Sapkota, et al. (2016). "Study on Effect of Adding Booster Runner in Conical Basin: Gravitational Water Vortex Power Plant: A Numerical and Experimental Approach." 4th IOE Graduate Conference.

Awards and Honors

- IOE Merit Fellowship, M.S. in Mechanical Engineering, Tribhuvan University, 2019
- Best Engineering Award and Panasonic Award, ABU Asia-Pacific Robot Contest, Thailand, 2016
- Small Research Development and Innovation Grant, University Grant Commission, Nepal, 2016
- Best Idea Award and Mabuchi Motor Award, ABU Asia-Pacific Robot Contest, Indonesia, 2015

Professional Memberships

- American Physical Society (APS), Member | 2024–Present
- Nepal Engineering Council, Registered Mechanical Engineer | 2018–Present